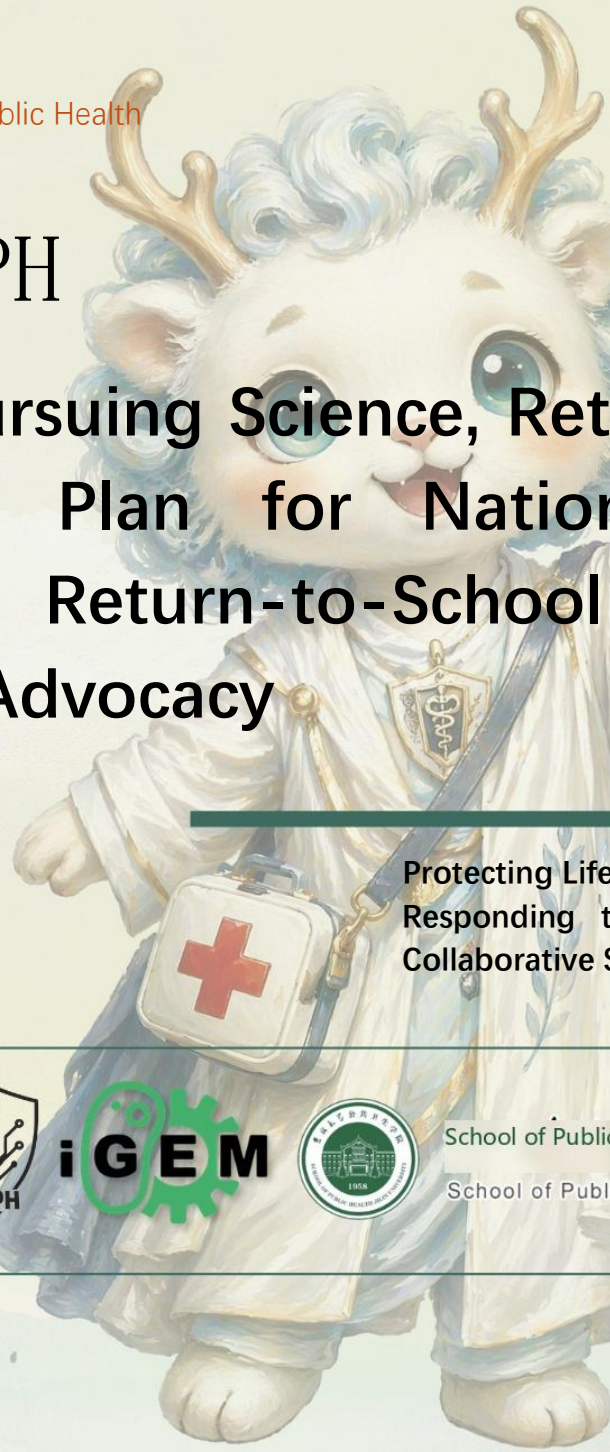
Life Protection – Public Health

JLU-SPH

2026 “Pursuing Science, Returning to Dreams” Plan for National High School Return-to-School Public Science Advocacy



Protecting Life with Scientific Insight
Responding to Public Health Through Collaborative Strength

School of Public Health, Jilin University

School of Public Health Jilin University

JILIN UNIVERSITY · SCHOOL OF PUBLIC HEALTH

I. Basic Information of the Activity

- **Activity Name:** "Pursuing Science and Technology Innovation, Returning to Chase Our Dreams"—National High School Alumni Public Science Advocacy Program
- **Event Theme:** Using university iGEM (International Genetically Engineered Machine) practice as a bridge to popularize synthetic biology and establish a pathway for high school students to explore scientific innovation and research.
- **Type of Activity:** ☐ Public Service ☑ Science Education & Popularization ☑ Cultural Promotion ☐ Community Service ☐ Other:
- **Organizer:** JLU- SPH Team

II. Purpose of the Activity

1. Core Objective: To utilize standardized and reproducible return-to-school public science lectures as a vehicle, leveraging university iGEM project experiences to promote basic knowledge of synthetic biology (synthetic biology) among high school students, popularize the iGEM scientific innovation competition, clarify feasible pathways for scientific innovation learning during the high school stage, and lower the barriers for frontier science and technology education for high school students.

2. Specific Objectives:

- Cognitive/Knowledge Perspective: Assist high school students in understanding fundamental concepts of synthetic biology, clarifying the essence of the iGEM competition, establishing basic approaches for initiating scientific innovation learning from scratch, and dispelling common misconceptions about scientific innovation competitions; simultaneously, help secondary school teachers grasp key points of support required for conducting scientific innovation activities.
- Behavioral/Skill Level: To guide students in developing scientific and technological innovation (kechuang) project-oriented thinking, learning to systematize their own exploration and planning in scientific innovation, and improving their ability to gather and critically appraise information on academic competitions and research practices.
- Emotional/Attitudinal Level: To stimulate high school students' enthusiasm for exploring the field of biology and science/technology innovation (kechuang), to help them overcome perceived barriers of high entry and unattainability in cutting-edge science, and to foster confidence in pursuing scientific innovation dreams.
- **3. Expected Outcomes:**
- Coverage: High school students and subject teachers from eight partner high schools nationwide
- Satisfaction Target: Over 90%

Ⅲ、Target Audience

- Core Group: High school students and their subject teachers from partner schools in non-tier-one cities who are interested in biology, scientific innovation (kechuang), and subject competitions.
- Audience Characteristics: Possess basic knowledge of high school biology; have the intention to

- explore science and technology innovation but lack channels or resources related to cutting-edge research and scientific competitions; require actionable and accessible guidance for science innovation learning; constitute a critical group for early science mentorship and competition planning.

IV. Date and Location of Activity

- Activity Date: Winter vacation, 2026
- Duration per Session: Approximately 40 minutes
- Location: Classrooms or multimedia lecture halls within eight partner high schools nationwide

V. Content and Procedure of the Activity

[Overview of Activity Procedure]

Phases	Time	Session Name	Specific Content	Personnel Responsible	Required Materials
Phase	Preparation Phase	Preliminary Needs Assessment	Assess students' knowledge base and teachers' practical needs for implementing science and technology innovation (kechuang); complete the standardized development of lecture content.	Project Lead, All Program Presenters	Survey questionnaire, initial draft of the presentation PPT, training checklist
Main Phase	Duration per session: approximately 40 minutes	Lecture Delivery	Team members deliver the presentation at partner high schools	All Program Presenters	Standardized presentation PPT, educational outreach materials
Main Phase	Duration per session: approximately 10 minutes	On-site Questionnaire Collection and Q&A Session	Distribute student feedback questionnaires, provide open, on-site communication, and address questions from students and teachers regarding scientific innovation and participation in iGEM.	All Program Presenters	Student feedback questionnaires, paper and pens
Concluding Phase	Program Wrap-up	Comprehensive Project Review	Aggregate all event data and anonymized survey responses from the 8 schools; analyze and optimize directions; archive the complete set of materials.	Project Lead	Anonymized survey data

[Detailed Description of Each Session]

Session 1: Preliminary Need Assessment and Standardized Content Development

- Core Content: Conduct research in target secondary schools to assess the current knowledge base and scientific innovation challenges of high school students, and collect actual needs of teachers for delivering scientific innovation education; develop a continuous learning module focused on "What is Synthetic Biology?", "How to Understand iGEM?", and "How to Start Scientific Innovation Learning," and complete initial drafts of related materials including presentation PPT, workflow chart, and presenter training checklist.
- Key Messaging: Bring authentic university iGEM practice back to secondary schools, making frontier scientific innovation more accessible, and helping students identify suitable starting points for scientific innovation.

Session 2: Lecture Delivery

- Core Content: Deliver presentations in eight partner secondary schools, combining theoretical explanations using the PPT, interactive classroom Q&A sessions, and distribution of educational outreach materials. Provide students with foundational knowledge of synthetic biology, deconstruct iGEM projects, explain actionable scientific innovation learning pathways for high school students, and enable them to discuss their learning challenges through on-site interaction.

Session 3: On-site Q&A and Questionnaire Collection

- Core Content: Facilitate open discussion after each presentation to respond to questions from students and teachers regarding scientific innovation learning, iGEM participation pathways, and learning obstacles; distribute and promptly collect student feedback questionnaires on-site.

VI. Personnel Structure and Assignment

- Project Lead (1 person): Responsible for overall project coordination, liaison with the eight partner high schools nationwide, management of the full project cycle (needs assessment – pilot phase – implementation – review), and resolution of cross-regional communication and emergent issues.
- Each school's lead presenter(s): Responsible for refining presentation content, iterating PPT development, designing survey questionnaires, mapping out pathways for scientific innovation learning, and reviewing all presentation materials; must be well-versed in iGEM projects and synthetic biology, with communication adapted to the cognitive level of high school students.
- On-site Teaching Assistants (several): Distribution of educational materials, maintenance of order on site, collection and return of questionnaires, and recording of on-site situations.
- Logistics & Documentation (2 persons): Coordination of materials, maintenance and updating of various template documents, photography and preservation of event media, organization and review of project materials, and completion of open-source documentation archiving.

VII. Inventory List

- Equipment:

- ○ Laptop Computer, Projector
- ○ Camera/High-definition Mobile Phone
- ● Teaching Materials:
- ○ Standardized presentation PPT
- ○ Speaker Training Checklist
- ○ Session preparation checklist
- ○ Student Feedback Questionnaire
- ○ Session Review Template
- ○ School Partnership Description Document
- ○ Preliminary Survey Questionnaire
- ● Logistics:
- ○ Science Popularization Promotional Materials (materials used for public science outreach)
- ○ Paper and Pens
- ○ Garbage Bags

VIII. Risk Contingency Plan

- Venue Risk: Temporary changes in equipment or facilities at partner secondary schools. Confirm multimedia equipment conditions with the school in advance, request the school to provide backup devices, and presenters should bring backup versions of the PPT.
- Execution Risk: High mobility of presenters between schools, leading to variability in presentation quality. Strictly implement presenter training checklist, conduct item-by-item review against the preparation checklist before each presentation, and record issues from each session in a retrospective template for timely iteration.

IX. Effect Evaluation

- Process Evaluation: Conduct on-site observation of classroom interaction during lectures, record student participation in each session, and preserve implementation evidence for every event.
- Satisfaction Evaluation: Collect students' feedback questionnaires in both paper and electronic forms, gather opinions from secondary school teachers and school management, and compile anonymized survey data.
- Session Review and Archiving: Consolidate session review records from all eight schools, identify feedback regarding obstacles to students' scientific innovation learning and paths for competition participation, prepare a comprehensive project summary, and complete the open-source release of all tools and materials.